IPPI Data Governance & Generation Suite

Automating the creation, quality, and lifecycle of Investigational Product Preparation Instructions

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Situation: IPPI Creation Today

On average 78 IPPIs drafted annually across DPAT authoring teams, CMS review teams, and multiple R&D functions

78
IPPIs drafted annually

25 hrs
First-draft effort per IPPI (3 SMEs, 80% of cases)

4 weeks
Draft-to-submission lifecycle (80% of IPPIs)

Current process: Manual, template-based, with multiple authors and rounds of review. IPPIs stored in TruVault (TV_TECH). Signed-off attribute forms filed since June 2023.

Current Process Challenges

The manual drafting process creates risk, inconsistency, and delays

- Manually labor-intensive task with no automation support
- Lack of consistency in structure of instructions across authors
- Missing details that can lead to **patient safety risks**
- Human errors in drafting dosage and instructions — serious consequences for patients and company
- Variability in wording and comprehensiveness across IPPIs
- Multiple rounds of revisions and reviews needed to catch errors
- Long cycle time — waiting on SMEs for first draft through back-and-forth revisions
- Drafting requires years of experience — new hires struggle without a streamlined process

Objectives: Minimize & Maximize

Creating a data governance and generation suite to transform IPPI creation

OBJECTIVE	TARGET
Minimize first-draft creation time	80% of IPPIs auto-generated
Minimize hours spent drafting	1,872 hands-on hours saved annually
Minimize draft-to-submission lifecycle	From 4 weeks → 1 week per IPPI
Maximize IPPI review rate	By reducing cycle time 75%

Cost avoidance: 9,360 hands-off hours per year. 1,560 FTE work days currently spent on the IPPI lifecycle reduced by 75%.

Execution: Main

Leveraging the DS Digital Twin generic code base for orchestration and automation

- Collect user impact information for work activities
- Obtain outcomes of historical IPPIs to train the system
- Create unified Functional Area / Management Operational Health dashboards
- Unify comprehension and writing style across all IPPIs
- Identify potential dosing, preparation, and administration errors
- Prevent dosing errors before instructions are published
- Lower adverse events caused by errors in dosage, preparation, and administration
- Optimize IPPI creation process — detect loops, missing points, and gaps

Execution: Historical & Diagnostic

Building on historical data to enable quality control and future-state automation

HISTORICAL — BUILDING FOR THE FUTURE

- Map the IPPI drafting process end-to-end
- Extract entities, methods, and formulas used in IPPI creation
- Find linkages between concepts
- Automated tools to extract concept chains and detect missing information

DIAGNOSTIC — QUALITY CONTROL

- Identify lack of details and wrong information in historical documents
- Rate comprehension of historical IPPIs
- Rate accuracy of historical IPPIs
- Link comprehension/accuracy scores to clinical study outcomes and adverse events
- Conformance: compare process model against historical IPPIs created

Execution: Predictive & Prescriptive

From automated drafting to continuous process enhancement

PREDICTIVE — QUALITY ASSURANCE

- Cover drafting for up to 80% of IPPIs yearly
- Drafts mimic highly comprehensive and accurate historical IPPIs
- Auto-generate first drafts that meet quality benchmarks

PRESCRIPTIVE — PROCESS ENHANCEMENT

- Enhance the mapped process to produce superior results
- Recommend optimal instruction patterns based on historical outcomes
- Continuously improve drafting templates from feedback loops

Expected Impact

Quantifiable cost reduction and cycle-time improvement across the IPPI lifecycle

1,872
Hands-on hours saved annually (first-draft automation)

9,360
Hands-off hours per year (cost avoidance)

75%
Reduction in FTE resource requirement

Lifecycle target: From 4 weeks → 1 week per IPPI. 1,560 FTE work days annually compressed into a fraction — freeing SME capacity for higher-value clinical decision-making.

Why This Matters

- Automates 80% of IPPI first-draft creation, eliminating manual bottleneck
- Prevents dosing and administration errors **before** instructions are published
- Unifies quality and comprehensiveness across all IPPIs
- Reduces draft-to-submission from 4 weeks to 1 week — 75% resource savings
- Provides a repeatable pattern for other document governance use cases

> **A data governance suite that protects patients, reduces cost, and frees SME capacity for higher-value work.**